

Borzblade PSX/PS2 Retro Render Toolkit for URP

A practical guide for setting up, tuning, and troubleshooting the Borzblade PSX/PS2 Retro Render Toolkit in Unity URP.

Overview

Borzblade PSX/PS2 Retro Render Toolkit for URP is a shader and renderer toolkit for building a PlayStation-inspired look in Unity. It combines retro material shaders with an optional fullscreen renderer feature.

The package root is:

```
Assets/Borzblade/RetroRenderToolkit/
```

Use the toolkit when you want control over vertex snap, texture pixelation, affine-style UV distortion, color reduction, ordered dither, cutout shadows, foliage wind, terrain treatment, simple retro water, outlines, and a final fullscreen retro pass.

What's Included

- URP lit retro shaders.
- Terrain shader support.
- Cutout and foliage shaders.
- Unlit cutout shader.
- Lit and unlit sprite shaders.
- Retro transparent water shader.
- Fullscreen retro renderer feature.
- Depth fog renderer feature.
- Screen-space outline renderer feature.
- Material outline shader.
- Editor toolkit window.
- Material presets.
- Material converter.
- Diagnostics tools.
- Documentation and sample materials.

Not Included

- Third-party character, environment, prop, or demo-scene assets shown in screenshots are not included unless specifically listed.
- The package focuses on shaders, renderer tools, presets, materials, and documentation.

Third-Party Assets / Screenshots

Promotional screenshots may use separate character, environment, prop, terrain, or demo assets to show the shaders in context. Those assets are not part of this package unless they are explicitly listed as included content.

Compatibility

Item	Value
Tested Unity version	6000.4.1f1
Tested URP version	17.4.0
Supported render pipeline	Universal Render Pipeline
Unsupported render pipelines	Built-in Render Pipeline and HDRP
Main menu path	Tools > Borzblade > Retro Render Toolkit
Toolkit window	Tools > Borzblade > Retro Render Toolkit > Open Toolkit

The shaders are authored for URP. If a material turns pink, check that the active render pipeline asset is URP and that the shader files still exist under the toolkit folder.

Folder Layout

Folder	Purpose
Art	Placeholder textures, icons, previews, and alpha card example textures.
Demo	Demo scenes, prefabs, and demo materials when present.
Documentation	Markdown docs and this manual.
Editor	Toolkit window, installer, material converter, material factory, and shader inspectors.
Materials	Sample materials and preset materials.
Runtime	Renderer feature and runtime settings classes.
Samples	Optional material, terrain, foliage, and renderer comparison sample folders.
Settings	Renderer feature materials, global profiles, and presets.
Shaders	Core includes, material shaders, terrain shaders, and fullscreen shader.

Quick Start

1. Open the Unity project.
2. Confirm the project is using a URP render pipeline asset.
3. Open Tools > Borzblade > Retro Render Toolkit > Open Toolkit.
4. In the Setup tab, press Install Retro Renderer Feature.
5. Press Create Default Materials.
6. Assign materials from:
 - Assets/Borzblade/RetroRenderToolkit/Materials/Hybrid
 - Assets/Borzblade/RetroRenderToolkit/Materials/Terrain
 - Assets/Borzblade/RetroRenderToolkit/Materials/Cutout
 - Assets/Borzblade/RetroRenderToolkit/Materials/Foliage
 - Assets/Borzblade/RetroRenderToolkit/Materials/UnlitCutout
 - Assets/Borzblade/RetroRenderToolkit/Materials/Water

- Assets/Borzblade/RetroRenderToolkit/Materials/Outline

- Assets/Borzblade/RetroRenderToolkit/Materials/Sprites

7. Optional: press **Install Depth Fog Feature** if you want proper scene fog from camera depth.

8. Optional: press **Install Screen Outline Feature** if you want screen-space outlines.

9. In the **Renderer** tab, choose a fullscreen preset such as **Clean PS2**, **Crunchy PSX**, **Horror PSX**, **Dark Fantasy**, **VHS / CRT**, **Mobile Fast**, or **Off/Neutral**.

10. Use the material inspector presets to tune each material type.

A good starting setup:

- **PSXPS2_CleanPS2** for characters, weapons, and important props.

- **PSXPS2_CrunchyPSX** for surfaces that can handle a rougher low-resolution look.

- **PSXPS2_TerrainHybrid** for Unity Terrain.

- **M_RetroFoliage_Grass**, **M_RetroFoliage_Bush**, or **M_RetroFoliage_TreeLeaves** for foliage cards.

- **M_RetroWater_CleanPS2** for transparent water planes.

- **M_RetroOutline_Black** when you need an inverted-hull mesh outline material.

- **M_RetroSpriteLit_Clean** or **M_RetroSpriteUnlit_Clean** for **SpriteRenderer** materials.

- **Mobile Fast** renderer and material presets for lower-end targets.

Shader Selection Guide

Shader	Unity shader name	Best use
Hybrid Lit	Borzblade/Retro Render Toolkit/PSX PS2 Hybrid Lit	Main lit material for characters, buildings, rocks, props, weapons, and most opaque meshes.
Terrain Lit	Borzblade/Retro Render Toolkit/PSX PS2 Terrain Lit	Unity Terrain material/template using TerrainLayers , normals, masks, holes, height blend, shadows, and URP terrain add-pass fallback.
Cutout Lit	Borzblade/Retro Render Toolkit/PSX PS2 Cutout Lit	Transparent alpha cards with lighting and shaped shadows, such as fences, grates, ropes, hair cards, beard cards, cloth tears, and leaves.
Foliage Lit	Borzblade/Retro Render Toolkit/PSX PS2 Foliage Lit	Transparent foliage material with wind, color variation, and top/bottom foliage tint. Best for grass, bushes, leaves, pine branches, and dead plants.
Unlit Cutout	Borzblade/Retro Render Toolkit/PSX PS2 Unlit Cutout	Low-cost transparent alpha cards, far foliage, sprite props, simple background cards, and mobile use.
Water	Borzblade/Retro Render Toolkit/PSX PS2 Water	Transparent retro water surfaces with stepped waves, foam, depth fade, fresnel tint, and retro texture/color controls.
Material Outline	Borzblade/Retro Render Toolkit/PSX PS2 Material Outline	Inverted-hull mesh outline material for selected characters, props, and readable silhouettes.
Sprite Lit	Borzblade/Retro Render Toolkit/PSX PS2 Sprite Lit	SpriteRenderer -friendly transparent sprite shader with URP lighting, tint, vertex color, retro geometry, and retro color controls.
Sprite Unlit	Borzblade/Retro Render Toolkit/PSX PS2 Sprite Unlit	Lower-cost SpriteRenderer -friendly transparent sprite shader for UI-like props, pickups, flat FX, and mobile sprites.
Fullscreen Global Pass	Hidden/Borzblade/Retro Render Toolkit/PSX PS2 Global Pass	Renderer feature shader for final-screen pixelation, color steps, dither, scanlines, vignette, contrast, saturation, and bleed.
Depth Fog Pass	Hidden/Borzblade/Retro Render Toolkit/PSX PS2 Depth Fog	Fullscreen depth-based fog with linear, exponential, stepped, and dithered fog options.

Shader	Unity shader name	Best use
Screen Outline Pass	Hidden/Borzblade/Retro Render Toolkit/PSX PS2 Screen Outline	Renderer feature shader for depth/normal based screen outlines.

Lighting Models

Hybrid, Cutout, Foliage, and Sprite Lit materials include a Lighting Model control.

Mode	Use it when
Standard URP	You want the existing URP-style lit behavior with normal maps, specular, shadows, and lightmapping where the shader supports it. This is the default for existing mesh materials.
Vertex Lit	You want cheaper, older-looking lighting that changes across vertices instead of every pixel. Good for props, sprites, and low-poly meshes.
Flat Lit	You want hard face lighting for a low-poly look. It ignores the smooth vertex normal in the final lit result.
Unlit	You want texture, tint, emission, alpha, material fog, and retro color controls without scene lighting.

These modes do not replace the separate unlit cutout shader. Use PSX PS2 Unlit Cutout or PSX PS2 Sprite Unlit when you want the cheapest dedicated unlit path.

Toolkit Window

Open the toolkit from:

Tools > Borzblade > Retro Render Toolkit > Open Toolkit

Tab	What it does
Setup	Shows pipeline status, finds renderer assets, installs renderer features, creates default materials, opens docs, and pings the package root.
Renderer	Selects URP renderer assets, installs/edits the retro renderer feature, edits depth fog, edits screen outlines, and applies renderer presets.
Materials	Creates default materials, cutout/foilage materials, water/outline materials, sprite materials, quick new materials, and lists toolkit materials.
Converter	Converts selected materials or all materials in a selected folder to a toolkit shader, including water, material outline, and sprite shaders.
Presets	Applies material presets to selected materials.
Terrain	Creates/selects the terrain material and applies terrain presets.
Cutout/Foliage	Creates cutout, foliage, and unlit cutout material sets.
Demo/Docs	Opens package folders and documentation.
Diagnostics	Checks Unity version, pipeline asset, selected materials, required folders, shader availability, renderer feature installation, and old folder leftovers.

Installing The Fullscreen Renderer Feature

The fullscreen pass is optional. Use it when you want the whole camera image to share the same retro treatment.

Install it from either place:

- Toolkit window: Setup > Install Retro Renderer Feature

- Menu item: Tools > Borzblade > Retro Render Toolkit > Install Retro Renderer Feature

The installer:

- Creates or updates
Assets/Borzblade/RetroRenderToolkit/Settings/GlobalProfiles/PSXPS2_GlobalPass.mat.
- Checks known renderer assets such as Assets/Settings/PC_Renderer.asset and Assets/Settings/Mobile_Renderer.asset.
- Searches all renderer data assets in the project.
- Adds PSX PS2 Retro Pass to each discovered URP renderer asset when it is missing.
- Assigns the global pass material to the renderer feature.
- Enables the feature and saves assets.

The runtime class is:

```
Borzblade.RetroRenderToolkit.RetroRendererFeature
```

A compatibility wrapper named PSXPS2RetroRendererFeature is also present so older renderer feature sub-assets continue to resolve.

The pass runs after URP post-processing. It fetches the color buffer and skips work when the feature is disabled or Intensity is zero.

Installing The Screen Outline Feature

The screen outline feature is optional. It uses the camera depth and normals textures, then writes outlines before the final fullscreen retro pass.

Install it from either place:

- Toolkit window: Setup > Install Screen Outline Feature
- Renderer tab: Screen Outline > Install Screen Outline Feature
- Menu item: Tools > Borzblade > Retro Render Toolkit > Install Screen Outline Feature

The runtime class is:

```
Borzblade.RetroRenderToolkit.PSXPS2ScreenOutlineRendererFeature
```

Use screen outlines for broad scene readability. Use the material outline shader when a specific mesh needs a controlled silhouette.

Installing The Depth Fog Feature

Depth fog is the main fog system for scenes. It samples the camera depth texture, computes fog from distance, and blends it before the final retro fullscreen pass. That means the fog also receives the final pixelation, color reduction, and screen treatment.

Install it from either place:

- Toolkit window: Setup > Install Depth Fog Feature
- Renderer tab: Depth Fog > Install Depth Fog Feature

- Menu item: Tools > Borzblade > Retro Render Toolkit > Install Depth Fog Feature

The runtime class is:

```
Borzblade.RetroRenderToolkit.PSXPS2DepthFogRendererFeature
```

The pass material is created at:

```
Assets/Borzblade/RetroRenderToolkit/Settings/GlobalProfiles/PSXPS2_DepthFog.mat
```

Use depth fog for the whole scene. Use material Retro Fog only when a specific shader needs a local stylized fade that should not depend on the camera depth texture.

Fullscreen Renderer Settings

Setting	Range	What it controls	Notes
Effect Enabled	On/off	Turns the renderer feature settings on or off.	Use this for quick comparisons.
Intensity	0 to 1	Blends between the original image and the processed image.	Start near 0.35 for subtle PS2. Use higher values for a stronger PSX look.
Pixel Scale	1 to 12	Size of screen-space pixel blocks.	1 is clean. 2 or 3 is visibly chunky.
Pixelation Mode	Scale / Fixed Vertical Resolution	Chooses between the normal pixel scale and a fixed vertical resolution target.	Use fixed vertical resolution when you want a consistent 240p-style or 360p-style output across different window sizes.
Fixed Vertical Resolution	64 to 1080	Target vertical resolution when fixed mode is enabled.	240 is a classic low-resolution baseline.
Color Steps	2 to 64	Final color quantization.	28 is balanced. 12 to 16 is much harsher.
Dither Strength	0 to 1	Ordered Bayer dither before color quantization.	Useful with low color steps. Disable it first when optimizing.
Dither Pattern Mode	Procedural / Texture	Uses the built-in procedural Bayer pattern or a user-supplied texture.	The package does not include third-party dither textures. Assign your own if needed.
Dither Pattern Texture	Texture	Custom dither pattern used when texture mode is selected.	Point-filtered grayscale textures work best.
Dither Pattern Scale	0.25 to 16	Scales the custom dither texture sampling.	Higher values make the pattern larger on screen.
Dither Threshold	0 to 1	Shifts the custom dither texture around the midpoint.	0.5 is neutral.
Scanline Strength	0 to 1	Darkens alternating scanline pattern.	Keep low unless the screen treatment is intentional.
Vignette Strength	0 to 1	Darkens image edges.	Useful for horror and darker scenes. Easy to overdo.
Saturation	0 to 2	Pushes color toward grayscale or stronger color.	1 is neutral.
Contrast	0 to 2	Adjusts final contrast.	Small changes are usually enough.
Color Bleed	0 to 1	Offsets color sampling horizontally.	Low values add a soft analog feel. High values look broken.
Color Tint	Color	Multiplies the final image.	Keep close to white for normal use.
Gamma	0.2 to 3	Adjusts midtone response.	Small changes are best.
Black Level	0 to 1	Raises the black cutoff before quantization.	Useful for darker horror looks.
Dither Scale	0.25 to 8	Scales the screen-space dither pattern.	1 matches the default Bayer pattern.

Setting	Range	What it controls	Notes
CRT Mask Strength	0 to 1	Adds a simple RGB stripe mask.	Keep low unless the camera is meant to look like a display capture.
Chromatic Offset	0 to 1	Separates red/blue sampling horizontally.	Low values are enough.
Noise Strength	0 to 1	Adds light temporal grain.	Use sparingly.
Horizontal Jitter	0 to 1	Adds per-line horizontal wobble.	Better for VHS/CRT presets than clean PS2.
Curvature	0 to 1	Bends screen UVs outward.	Optional; leave off for a plain game-camera look.
Global Fog Enabled	On/off	Adds a simple fullscreen fog tint before final quantization.	Kept for older presets and quick tinting. Prefer Depth Fog for normal scene fog.
Global Fog Color	Color	Fog tint color.	Alpha also controls tint strength.
Global Fog Intensity	0 to 1	Blends the fog tint over the final image.	Keep low for subtle haze.

Depth Fog Settings

Depth fog reads scene depth, so it follows actual camera distance instead of guessing per material. It works best when the active URP renderer has depth texture support available.

Setting	What it controls	Notes
Effect Enabled	Turns the depth fog pass on or off.	The renderer feature can also be disabled from the renderer asset.
Pass Event	When the pass runs in URP.	Default is BeforeRenderingPostProcessing, before the final retro pass.
Fog Color	Fog tint and alpha strength.	Alpha multiplies the final fog amount.
Intensity	Overall blend strength.	Start around 0.25 to 0.45.
Start Distance	Distance where fog begins.	Use scene scale units.
End Distance	Distance where linear fog reaches full strength.	Also used to normalize stepped fog.
Density	Exponential fog density.	Used by exponential blend modes.
Blend Mode	Linear, Exponential, or Exponential Squared.	Linear is predictable. Exponential modes feel denser near the far range.
Steps	Quantizes the fog amount.	Higher values are smoother. 0 or 1 disables stepping.
Dither Strength	Breaks up fog bands with ordered dither.	Useful when Steps is low.
Dither Scale	Scale of the fog dither pattern.	1 is the default 4x4 Bayer scale.
Affect Sky	Applies fog to sky/background depth.	Leave off when the skybox should stay clear.

Renderer Presets

Preset	Settings	Use case
Clean PS2	Intensity 0.35, Pixel Scale 1, Color Steps 40, Dither 0.04, Scanlines 0.02, Vignette 0.06, Saturation 1.03, Contrast 1.03, Bleed 0.02	Subtle final pass for a cleaner low-poly look.
Crunchy PSX	Intensity 0.85, Pixel Scale 3, Color Steps 12, Dither 0.34, Scanlines 0.15, Vignette 0.18, Saturation 0.95, Contrast 1.14, Bleed 0.14	Strong screen-space PSX treatment.
Horror PSX	Intensity 0.78, Pixel Scale 2, Color Steps 16, Dither 0.28, Scanlines 0.12, Vignette 0.38, Saturation 0.78, Contrast 1.22, Bleed 0.10	Darker, lower-color scenes.
Dark Fantasy	Intensity 0.60, Pixel Scale 1.5, Color Steps 20, Dither 0.18, Scanlines 0.06, Vignette 0.26, Saturation 0.90, Contrast 1.12, Bleed 0.08	Moody fantasy scenes.

Preset	Settings	Use case
Mobile Fast	Intensity 0.28, Pixel Scale 1, Color Steps 28, Dither 0, Scanlines 0, Vignette 0.05, Saturation 1, Contrast 1, Bleed 0	Low-cost mild treatment.
Off/Neutral	Intensity 0, Pixel Scale 1, Color Steps 64, Dither 0, Scanlines 0, Vignette 0, Saturation 1, Contrast 1, Bleed 0	Disables the fullscreen effect.
VHS / CRT	Moderate pixel scale, scanlines, color bleed, mask, chromatic offset, noise, line jitter, and slight curvature.	A stronger analog-display pass.

Material Presets

Presets are starting points. Apply one, then tune the material in context with your lighting, textures, camera distance, and target platform.

Preset group	Presets
Hybrid Lit	Hybrid High, Crunchy PSX, Clean PS2, Mobile Fast
Terrain Lit	Mountain Path / Hybrid Terrain, Crunchy PSX Terrain, Clean PS2 Terrain, Mobile Terrain
Cutout Lit	Clean Cutout, Crunchy PSX Cutout, Dither Fade Cutout, Hair / Beard Card, Fence / Gate, Mobile Cutout
Foliage Lit	Grass Card, Bush Leaves, Tree Leaves, Pine Branch, Dead Bush, Dark Fantasy, Mobile Fast
Unlit Cutout	Distant Foliage, Sprite Prop, Dither Fade Card, Mobile Fast
Water	Clean PS2, Crunchy PSX, Swamp, Mobile Fast
Material Outline	Black, Warm, Crunchy

The Presets tab also has a My Presets section. It can save the selected material's geometry, texture modifier, color, lighting, wind, fog, and retro renderer-related shader values as a RetroMaterialPreset asset under:

```
Assets/Borzblade/RetroRenderToolkit/Settings/UserPresets/
```

User presets do not store the base texture or base color. That keeps them useful across different props, foliage cards, and characters. You can load a saved preset onto selected materials, delete it with confirmation, or export/import it as JSON.

Hybrid Lit Shader

Use PSX PS2 Hybrid Lit for most lit mesh assets.

Good uses:

- Characters and enemies.
- Weapons, armor, tools, pickups, crates, doors, walls, roofs, stones, ruins, props, and architecture.
- Opaque materials that need shadows, normal maps, emission, rim light, PS2-style specular, and retro texture/color controls.

Use another shader for these cases:

- Unity Terrain: use PSX PS2 Terrain Lit.
- Alpha-card foliage: use PSX PS2 Foliage Lit or PSX PS2 Cutout Lit.
- Cheap distant cards or sprite props: use PSX PS2 Unlit Cutout.

- Transparent water planes: use PSX PS2 Water.
- Mesh outlines: use PSX PS2 Material Outline or the screen outline renderer feature.

Hybrid Lit Inspector Sections

Section	Important controls
Surface	Base Map, Base Color, Normal Map, Normal Scale, Alpha Clip Threshold, Surface Type, Blend Mode, Cull.
Geometry Modifiers	Vertex Snap, Vertex Snap Strength, Vertex Snap Resolution, Distance Fade, Fade Start/End, Snap Seam Reduction, optional baked snap anchors, Vertex Wobble, Wobble Strength, Speed, Scale.
Texture Modifiers	UV Pixelation, UV Pixel Strength, UV Pixel Resolution, UV Pixel Aspect, Mip Bias, Affine Warp, Affine Strength.
Color Modifiers	Posterize, Posterize Steps, Palette Strength, Palette Steps, Dither, Dither Strength, Dither Scale.
Lighting Modifiers	Shadow/Light Bands, Band Strength, Rim, Rim Color, Rim Intensity, Rim Power, Specular Mode, PS2 specular settings.
Retro Fog	Per-material fog toggle, color, start/end, density, steps, and linear/exponential blend mode.
Advanced	Render queue, alpha state, receive shadows, and low-level render state controls.

Hybrid Lit Presets

Preset	Main behavior
Hybrid High	Balanced default. Vertex snap on, mild wobble, UV pixelation, posterization, dither, rim light, and medium specular.
Crunchy PSX	Strong vertex snap, low snap resolution, higher wobble, heavy UV pixelation, affine warp, low posterize steps, stronger palette compression, dither, and light bands.
Clean PS2	Vertex snap and wobble off, high UV resolution, subtle posterization, no dither, rim enabled, stronger glossy specular.
Mobile Fast	Lower-cost settings. Moderate vertex snap, wobble off, affine off, no dither, rim off, lower specular.

Cutout Lit Shader

Use PSX PS2 Cutout Lit for transparent alpha-card geometry that should receive lighting and cast shaped shadows.

Good uses:

- Fences and grates.
- Hair and beard cards.
- Torn cloth and banners.
- Ropes, branches, leaves, and card details that are not using the foliage shader.
- Any alpha texture where a square shadow would be wrong.

The visible pass uses transparent blending, while the depth and shadow passes still use alpha cutoff. This keeps soft texture alpha visible in URP while preserving shaped card silhouettes for shadows and depth.

Cutout Settings

Setting	What it does
Alpha Cutoff	Controls clipping in depth and normal passes. Higher values remove more pixels from depth silhouettes.
Shadow Cutoff	Controls clipping in the shadow pass. Raise it if shadows are too thick. Lower it if shadows disappear.
Dithered Cutout Fade	Fades card visibility by distance or camera fade. The forward pass fades alpha; depth and shadow passes use dithered coverage.
Dither Fade Amount	Maximum dither fade amount.
Dither Fade Start/End	Distance range for distance-based dither fade.
Camera Fade	Dither-fades cards close to the camera. Useful for grass or leaves near the view.
Camera Fade Distance	Distance where close-camera fade is strongest.
Distance Fade	Enables distance-based dither fade using the fade start/end values.
Backface Tint	Tints reverse faces so two-sided cards read less flat.
Two Sided	Disables culling so cards are visible from both sides. Useful for leaves and hair.

Cutout Presets

Preset	Use case
Clean Cutout	General card material with subtle retro treatment.
Crunchy PSX Cutout	Stronger vertex snap, affine warp, pixelation, posterization, palette compression, dither, and light bands.
Dither Fade Cutout	Distance fade for cards that should disappear gradually.
Hair / Beard Card	Lower cutoff, tuned shadow cutoff, two-sided rendering, lower rim/specular.
Fence / Grate	Higher cutoff, backface culling, moderate vertex snap, good for hard alpha shapes.
Mobile Cutout	Lower-cost dither, rim, and specular settings.

Foliage Lit Shader

Use PSX PS2 Foliage Lit for grass, bushes, tree leaves, pine branches, and plant cards. It starts from the transparent alpha-card workflow and adds wind and foliage color controls.

Foliage / Wind Settings

Setting	What it does
Wind Enabled	Enables vertex movement for foliage.
Wind Strength	Multiplier for the local jiggle amount. Keep it low for tree leaves and moderate for grass.
Jiggle Amplitude (_WindDistance)	Maximum local flutter amount. It no longer sweeps the whole card sideways. UV and vertex masks can reduce it per vertex.
Wind Speed	Animation speed. Very high values can shimmer with vertex snap.
Wind Scale	Spatial frequency of wind variation.
Wind Direction	Local tangent/bitangent bias for the flutter direction.
Vertex Color Alpha Wind Mask	Uses vertex alpha as a wind mask. Paint darker alpha near stems to anchor them.
UV Height Wind Mask	Uses UV height so the top of a card moves more than the bottom. Useful for grass.
Top/Bottom Gradient	Multiplies color by a vertical foliage gradient.
Top Color and Bottom Color	Gradient colors for foliage tinting.
Color Variation	Adds per-position color variation to reduce repetition.
Hue/Saturation/Brightness Variation	Controls the variation channels.

Setting	What it does
Mobile Lighting	Simplifies foliage lighting for lower cost.

Foliage Presets

Preset	Use case
Grass Card	Stronger wind, lower cutoff, green top/bottom gradient.
Bush Leaves	Balanced bush preset with moderate wind and green variation.
Tree Leaves	Lower wind, smaller wind scale, lower specular/rim.
Pine Branch	Higher cutoff and darker green palette.
Dead Bush	Brown dry plant coloring and more saturation variation.
Dark Fantasy	Blue-gray fantasy foliage with stronger posterization, dither, and light bands.
Mobile Fast	Lower wind, no foliage variation, mobile lighting on, dither and rim off.

Unlit Cutout Shader

Use PSX PS2 Unlit Cutout when lighting is not needed or the object is far from the camera.

Good uses:

- Distant tree cards.
- Sprite-like prop cards.
- Background card silhouettes.
- Cheap mobile foliage.
- Simple cards where lighting cost is not worth it.

This shader supports Base Map, Base Color, transparent alpha blending, Alpha Cutoff for depth, Shadow Cutoff, Emission Color, Emission Map, UV pixelation, posterization, palette compression, ordered dither, dithered distance fade, two-sided state, ShadowCaster, and DepthOnly passes.

It does not use normal maps, PS2 specular, rim lighting, or foliage wind.

Sprite Shaders

Use the sprite shaders on Unity SpriteRenderer objects when you want sprite texture assignment, renderer tint, and sprite vertex color support.

Shader	Use case
PSX PS2 Sprite Lit	Sprites that should respond to URP scene lighting. It supports the shared Lighting Model control, normal map, specular, emission, retro geometry, UV pixelation, affine modes, posterization, dither, material fog, and vertex draw distance.
PSX PS2 Sprite Unlit	Sprites that should stay flat and cheap. It keeps tint, emission, retro geometry, UV pixelation, affine modes, posterization, dither, material fog, and vertex draw distance.

Create default sprite materials from:

Tools > Borzblade > Retro Render Toolkit > Create Sprite Materials

The generated materials are stored in:

Assets/Borzblade/RetroRenderToolkit/Materials/Sprites

Sprite shaders use transparent blending. They follow Unity's normal SpriteRenderer sorting rules, so sorting layer, order in layer, material queue offset, and camera transparency sorting still matter.

Water Shader

Use PSX PS2 Water for simple transparent water planes, ponds, streams, pools, and stylized shoreline surfaces.

Main controls:

Setting	What it does
Surface Tint, Shallow Color, Deep Color	Sets the base water palette. Depth fade blends between shallow and deep colors when the camera depth texture is available.
Alpha	Controls transparent blend strength.
Wave / Foam Noise	Optional noise texture for wave and foam variation.
Vertex Wave Strength, Wave Speed, Wave Scale, Wave Steps	Controls stepped water motion. Lower steps look more retro.
Depth Fade and Depth Fade Distance	Uses scene depth for water color depth. Requires camera depth texture support in URP.
Foam Color, Foam Distance, Foam Strength	Adds simple edge foam from depth difference and noise.
Fresnel Color, Fresnel Intensity, Fresnel Power	Adds a rim tint at grazing angles.
Retro Fog	Optional per-material fog for stylized water tinting. Use depth fog for the main scene haze.

Water uses transparent blending, so it can still be affected by normal Unity transparency sorting. Keep large water surfaces simple, avoid stacking many transparent planes, and use the mobile preset when depth fade is not needed.

Outline Features

There are two outline options.

Option	Use it when
Screen outline renderer feature	You want camera-wide outlines from scene depth and normals. Install it from the Setup or Renderer tab.
Material outline shader	You want an intentional silhouette on a specific mesh. Assign PSX PS2 Material Outline to a duplicate mesh or outline submesh.

Screen outline settings:

Setting	What it does
Pass Event	Controls where the outline pass runs. The default is before post-processing so the final retro pass also affects the outlines.
Outline Color	Final outline color.
Intensity	Overall strength.
Thickness	Neighbor sampling distance in screen pixels.
Depth Sensitivity	How strongly depth changes create outlines.
Normal Sensitivity	How strongly normal changes create outlines.
Distance Fade Start/End	Fades outlines over distance.

Setting	What it does
Blend	Blends the outline color over the scene.

Material outline settings:

Setting	What it does
Outline Color	Mesh outline color and alpha.
Outline Thickness	Object-space normal extrusion.
Distance Fade	Softens the outline over distance.
Vertex Snap and Dither	Optional retro treatment for the outline mesh.

Terrain Lit Shader

Use PSX PS2 Terrain Lit as the material/template for Unity Terrain.

Setup:

1. Open Tools > Borzblade > Retro Render Toolkit > Open Toolkit.
2. Go to the Terrain tab.
3. Press Create Terrain Material.
4. Select your Terrain object.
5. In the Terrain component, assign Assets/Borzblade/RetroRenderToolkit/Materials/Terrain/PSXPS2_TerrainHybrid.mat to the material/template slot.
6. Continue painting TerrainLayers with Unity's normal Terrain tools.

TerrainLayers still handle:

- Albedo textures.
- Normal maps.
- Mask maps.
- Tiling and offsets.
- Metallic and smoothness values.
- Height data.

The retro terrain material adds vertex snap, optional terrain vertex wobble, TerrainLayer UV pixelation, posterization, palette compression, ordered dither, optional light bands, optional material fog, terrain holes support, and the required depth, shadow, normal, scene selection, and meta passes. When Unity needs extra terrain passes for more than four painted layers, the shader uses URP's built-in terrain add pass so the main terrain shader imports reliably.

Terrain Presets

Preset	Main behavior
Mountain Path/Hybrid Terrain	Balanced default. Vertex snap on, UV pixelation on, posterization and dither on, light bands off.
Crunchy PSX Terrain	Stronger snap, wobble on, low UV resolution, low posterize steps, stronger palette/dither, and light bands on.

Preset	Main behavior
Clean PS2 Terrain	Subtle snap, high snap resolution, very low UV pixelation, higher posterize steps, low dither, no light bands.
Mobile Terrain	Vertex snap on, TerrainLayer UV pixelation off, dither off, light bands off, instanced per-pixel normal off.

Unity Terrain note: reliable height blending can break when a terrain needs extra add passes for more than four layers. If height blending looks wrong, reduce the number of active painted layers per terrain tile or turn height blending off.

Material Parameter Reference

Surface And Texture Inputs

Property	Applies to	Meaning
Base Map / _BaseMap	Mesh material shaders	Main texture. Alpha is used by cutout shaders.
Sprite Texture / _MainTex	Sprite shaders	SpriteRenderer texture. Supports sprite tint and renderer vertex color.
Base Color / _BaseColor	Mesh material shaders	Multiplies the base texture. Alpha also affects cutout opacity.
Tint / _Color	Sprite shaders	Multiplies the sprite texture and SpriteRenderer color.
Normal Map / _BumpMap	Lit shaders	Tangent-space normal map.
Normal Scale / _BumpScale	Lit shaders	Strength of the normal map.
Emission Color / _EmissionColor	Hybrid, Cutout, Foliage, Unlit Cutout, Sprite, Water	HDR emission multiplier where supported.
Emission Map / _EmissionMap	Hybrid, Cutout, Foliage, Unlit Cutout, Sprite	Emission texture.
Mip Bias / _MipBias	Hybrid, Cutout, Foliage, Unlit Cutout, Water, Sprite	Biases texture mip level. Negative values sharpen, positive values soften.

Geometry Controls

Property	Meaning	Typical values
Vertex Snap Enabled	Quantizes clip-space vertex positions to a low-resolution grid.	On for PSX, off or subtle for clean PS2.
Vertex Snap Strength	Blend amount between normal geometry and snapped geometry.	0.05 to 0.35 subtle, 0.55+ strong.
Vertex Snap Resolution	Grid resolution used for snapping. Lower means chunkier movement.	128 to 240 PSX, 320+ cleaner.
Vertex Snap Space	Chooses screen-space snap or view/world-style snap.	Screen is the classic default. View/World feels more like object-space jitter and can be calmer on some meshes.
Vertex Snap Distance Fade	Fades snap by camera distance.	Use higher values to reduce distant crawl.
Snap Fade Start/End	Distance range for snap fade.	Adjust to your scene scale.
Snap Seam Reduction	Reduces visible face gaps caused by heavy clip-space snapping.	0 is rawer. 0.15 to 0.35 is usually enough.
Use Baked Snap Anchors	Uses UV4 snap-anchor data baked by the toolkit.	Optional. Use it for meshes that crack at shared edges.
Vertex Draw Distance	Clips geometry past a material-controlled distance when enabled.	Off by default. Use for props that should disappear at a retro draw distance.
Vertex Draw Distance Fade	Dither-fades the cutoff instead of using a hard pop.	0 gives a hard cutoff. Small fade widths are usually enough.
Vertex Wobble Enabled	Adds animated normal-direction wobble.	Use carefully on characters and foliage.
Vertex Wobble Strength	Amount of wobble.	0.04 to 0.12 subtle, 0.25+ stylized.
Vertex Wobble Speed	Animation speed.	Keep moderate to avoid shimmer.

Property	Meaning	Typical values
Vertex Wobble Scale	Spatial frequency of wobble.	Higher values vary more across the mesh.

Texture Controls

Property	Meaning	Notes
UV Pixelation Enabled	Quantizes UV coordinates before sampling.	Use for chunky texture sampling.
UV Pixelation Strength	Blend amount between original and pixelated UVs.	Start around 0.08 to 0.22.
UV Pixelation Resolution	Virtual texture grid resolution.	Lower values such as 128 look rougher.
UV Pixelation Aspect	Non-square UV pixel aspect multiplier.	Leave 1 unless you want stretched texel blocks.
Affine Warp Enabled	Simulates affine UV interpolation by carrying a clip-W UV pair from vertex to fragment.	Strongest on large polygons viewed at grazing angles.
Affine Warp Strength	Blend between perspective-correct UVs and affine UVs.	0 is normal. 1 is full affine.
Affine Mode	Chooses Stable or Classic affine behavior.	Stable is the default. Classic is stronger and closer to older PSX-style texture swimming.

Color Controls

Property	Meaning	Notes
Posterize Enabled	Reduces color precision per material.	Useful even with fullscreen color steps.
Posterize Steps	Number of color steps.	10 to 16 crunchy, 24 balanced, 36+ clean.
Palette Strength	Blends toward a palette-compressed color.	Higher values create stronger color grouping.
Palette Steps	Number of palette steps.	16 crunchy, 32 balanced.
Dither Enabled	Adds ordered Bayer dither.	Helps banding and adds retro texture.
Dither Strength	Amount of ordered dither.	Keep lower on mobile or high-resolution displays.
Dither Scale	Screen-space dither scale.	1 is the default.

Lighting Controls

Property	Meaning	Notes
Specular Color	Color of PS2-style highlight.	Neutral gray/white fits most materials.
Smoothness	Smoothness used by lit shading.	Higher values make tighter highlights.
PS2 Specular Intensity	Extra main-light specular intensity.	Clean PS2 can use stronger values than PSX.
PS2 Specular Power	Specular exponent.	Higher values make smaller highlights.
Specular Mode	Chooses Legacy Per Pixel, PS1 Off, or PS2 Vertex specular.	Existing materials keep Legacy by default. PSX presets disable specular. PS2 presets use vertex specular.
Lighting Model	Chooses Standard URP, Vertex Lit, Flat Lit, or Unlit.	Hybrid, Cutout, Foliage, and Sprite Lit support it. Standard URP is the compatibility default for mesh shaders.
Shadow/Light Bands	Quantizes perceived lighting intensity.	Use 4 to 6 for stylized banding.
Band Strength	Blend amount for lighting bands.	Avoid high values on faces unless that is the intended style.
Rim Enabled	Adds view-dependent rim light.	Good for characters and silhouettes.
Rim Color	Rim light color.	Use scene lighting as a guide.
Rim Intensity	Rim brightness.	Keep subtle for clean PS2.
Rim Power	Rim falloff.	Higher values make a thinner rim.

Retro Fog Controls

Hybrid Lit, Cutout Lit, Foliage Lit, Unlit Cutout, Terrain Lit, and Water include optional per-material retro fog. It is calculated from camera distance, then quantized by Fog Steps. Use Linear for direct start/end fog. Use Exponential when you want fog to build up more gradually from the camera.

Material fog only changes color. It does not change alpha, render queue, or transparency sorting.

For normal whole-scene fog, install PSXPS2DepthFogRendererFeature instead. Depth fog uses the camera depth texture, so it handles mixed objects more consistently than per-material fog.

Converting Existing Materials

Open the converter here:

Tools > Borzblade > Retro Render Toolkit > Open Toolkit > Converter

Option	Meaning
Target Shader	Converts to Hybrid, Terrain, Cutout, Foliage, Unlit Cutout, Water, or Material Outline.
Create Copies	Copies materials instead of overwriting originals. Recommended for first tests.
Preserve Properties	Preserves common base texture, color, normal, emission, cutoff, cull, alpha clip, smoothness, and queue values.
Backup If Overwriting	Creates a backup material before overwriting an original.
Apply Preset	Applies a toolkit preset after conversion.
Dry Run	Lists what would be converted before doing it.

Suggested workflow:

1. Select one or more materials in the Project window.
2. Open the Converter tab.
3. Keep Create Copies on for the first pass.
4. Keep Preserve Properties on.
5. Pick a target shader and preset.
6. Press Dry Run.
7. If the preview is correct, press Convert Selected Materials.
8. Inspect the new material copies in a test scene.
9. Only overwrite originals after the copies are verified.

For folder conversion, select a folder in the Project window and press Convert Materials In Selected Folder.

Practical Recipes

Clean PS2 Scene

1. Install the renderer feature.
2. Apply renderer preset Clean PS2.

3. Use PSXPS2_CleanPS2 for characters and important props.
4. Use Clean PS2 Terrain for terrain.
5. Keep vertex snap off or low on hero characters.
6. Use UV pixelation around 0.08 to 0.18.
7. Keep dither low or disabled.
8. Use higher posterize steps such as 36 or 40.

Crunchy PSX Scene

1. Apply renderer preset Crunchy PSX.
2. Use PSXPS2_CrunchyPSX on props and surfaces that can handle stronger artifacts.
3. Use Crunchy PSX Terrain.
4. Enable affine warp on surfaces that can tolerate distortion.
5. Keep faces, UI-facing items, and key weapons less extreme if readability matters.
6. Use cutout shaders for alpha cards so shadows match silhouettes.

Dark Fantasy / Horror Scene

1. Apply renderer preset Dark Fantasy or Horror PSX.
2. Lower saturation through the renderer or through material colors.
3. Raise vignette carefully.
4. Use light bands on terrain and non-character props.
5. Keep character rim light subtle so silhouettes stay readable.
6. If enemies or UI become hard to see, reduce vignette and contrast first.

Foliage Cards

1. Use PSX PS2 Foliage Lit.
2. Keep Two Sided enabled for grass and leaves.
3. Assign a texture with a real alpha channel to Base Map.
4. Tune texture alpha and Base Color alpha until the visible card shape is clean.
5. Tune Shadow Cutoff separately so shadows are not too full or too thin.
6. Enable wind, but keep vertex snap and wobble subtle.
7. Use UV Height Wind Mask for grass cards so bases remain stable.
8. Use Vertex Color Alpha Wind Mask if the mesh has vertex colors painted.

Fences, Hair, And Grates

1. Use PSX PS2 Cutout Lit.
2. Use the texture alpha for the visible card shape.
3. For hair and beard cards, use lower cutoff for depth/shadow silhouettes and two-sided rendering.
4. For fences and grates, use a higher cutoff and backface culling when the model has real thickness.

5. Check shadows in-scene. Adjust `Shadow Cutoff` when only the shadow needs correction.

Mobile / Low Cost

1. Use renderer preset `Mobile Fast`.
2. Use material preset `Mobile Fast`.
3. Prefer `Unlit Cutout` for distant cards.
4. Disable dither where possible.
5. Keep screen pixel scale at 1.
6. Avoid heavy vertex wobble.
7. Disable terrain instanced per-pixel normals if terrain cost is high.
8. Use fewer overlapping foliage cards.

Texture Import Tips

- Smaller source textures usually fit the style better.
- Try point filtering on textures that are meant to stay crisp.
- Keep alpha textures clean for transparent card materials.
- Use `Alpha Cutoff` and `Shadow Cutoff` to tighten depth and shadow silhouettes when soft alpha edges look too broad.
- Use `Mip Bias` if mip levels make surfaces too blurry or too noisy.
- Use normal maps lightly on crunchy PSX materials. Too much surface detail can fight the low-resolution look.

Troubleshooting

Materials Turn Pink

Check:

1. The active pipeline is URP.
2. Shader files exist under `Assets/Borzblade/RetroRenderToolkit/Shaders`.
3. The package root has been reimported.
4. The toolkit `Diagnostics` tab reports the shaders as available.
5. `PSXPS2_GlobalPass.mat` uses `Hidden/Borzblade/Retro Render Toolkit/PSX PS2 Global Pass` if only the fullscreen pass is failing.

Fullscreen Effect Does Not Appear

Check:

1. Open the toolkit `Renderer` tab.
2. Confirm your active URP renderer asset appears in the list.
3. Press `Install Feature`.
4. Select the renderer asset and confirm `PSX PS2 Retro Pass` is present.

5. Confirm Effect Enabled is on.
6. Confirm Intensity is greater than zero.
7. Make sure the camera uses the renderer that has the feature.

Foliage Appears As Squares

Use Cutout Lit, Foliage Lit, or Unlit Cutout. Assign a base texture with alpha. The visible pass blends transparency; use Alpha Cutoff for depth silhouettes.

If the object still appears square, check that the imported texture actually has alpha and that the material uses the correct texture.

Shadows Are Square

Use a cutout shader so the ShadowCaster pass clips alpha. Then tune Shadow Cutoff.

Transparent Sorting Looks Wrong

Cutout, foliage, and unlit cutout now use transparent blending in the visible pass. If sorting looks wrong, reduce overlapping cards, split dense card clusters, or adjust Queue Offset on specific materials.

Too Much Geometry Crawling

Lower:

- Vertex Snap Strength
- Vertex Wobble Strength
- Fullscreen Pixel Scale
- Affine Warp Strength

Raise:

- Vertex Snap Resolution
- Posterize Steps

For important hero characters, use Clean PS2 or a custom material with vertex snap disabled.

If faces separate at shared edges, lower Vertex Snap Strength first. If the mesh still cracks, raise Snap Seam Reduction. For problem meshes, use Tools > Borzblade > Retro Render Toolkit > Bake Snap Anchors For Selection, then enable Use Baked Snap Anchors on the material. The baker creates duplicate mesh assets under Assets/Borzblade/RetroRenderToolkit/Generated/SnapAnchors/.

Scene Is Too Dark

Lower:

- Renderer Vignette Strength
- Renderer Contrast
- Material Shadow/Light Bands

- Material Band Strength

Raise scene lighting or material Base Color if the scene is still too dark.

Foliage Shimmers Or Moves Too Far

Lower:

- Wind Strength
- Wind Speed
- Vertex Snap Strength
- Vertex Wobble Strength
- Dither Strength

Also check that the texture alpha edge is clean.

Jiggle Amplitude (`_WindDistance`) is the hard maximum local flutter amount. If changing it has no visible effect, check that Wind Enabled is on, Wind Strength is above zero, and the UV height or vertex color alpha wind mask is not keeping the card anchored. The foliage shader is designed to keep the card anchored and jiggle leaves in place.

Depth Fog Does Not Appear

Check that `PSXPS2DepthFogRendererFeature` is installed on the renderer used by the active camera. The pass needs scene depth. If the fog looks too smooth, raise Steps or Dither Strength. If the sky becomes fogged, turn off Affect Sky.

Textures Look Too Smooth

PSX-style materials usually need point-filtered textures. The material inspectors warn when an assigned texture is not using Point filtering and include a button to update the texture importer.

Build And Source Control Notes

- Keep materials that reference toolkit shaders in your project so Unity includes the shaders in builds.
- Keep `PSXPS2_GlobalPass.mat` if you use the fullscreen renderer feature.
- Commit Unity .meta files for documentation and material assets.
- Keep the compatibility wrapper `PSXPS2RetroRendererFeature` unless you also migrate existing renderer feature sub-assets.
- Runtime code uses the namespace `Borzblade.RetroRenderToolkit`.
- Editor code uses the namespace `Borzblade.RetroRenderToolkit.Editor`.

File Reference

Important files:

- `Assets/Borzblade/RetroRenderToolkit/Runtime/Rendering/PSXPS2RetroRendererFeature.cs`
- `Assets/Borzblade/RetroRenderToolkit/Runtime/Rendering/PSXPS2DepthFogRendererFeature.cs`
- `Assets/Borzblade/RetroRenderToolkit/Runtime/Rendering/PSXPS2ScreenOutlineRendererFeature.cs`

- Assets/Borzblade/RetroRenderToolkit/Shaders/Fullscreen/PSXPS2GlobalPass.shader
- Assets/Borzblade/RetroRenderToolkit/Shaders/Fullscreen/PSXPS2DepthFog.shader
- Assets/Borzblade/RetroRenderToolkit/Shaders/Hybrid/PSXPS2HybridLit.shader
- Assets/Borzblade/RetroRenderToolkit/Shaders/Terrain/PSXPS2TerrainLit.shader
- Assets/Borzblade/RetroRenderToolkit/Shaders/Terrain/PSXPS2TerrainAddPass.shader
- Assets/Borzblade/RetroRenderToolkit/Shaders/Cutout/PSXPS2CutoutLit.shader
- Assets/Borzblade/RetroRenderToolkit/Shaders/Foliage/PSXPS2FoliageLit.shader
- Assets/Borzblade/RetroRenderToolkit/Shaders/Unlit/PSXPS2UnlitCutout.shader
- Assets/Borzblade/RetroRenderToolkit/Shaders/Water/PSXPS2Water.shader
- Assets/Borzblade/RetroRenderToolkit/Shaders/Sprites/PSXPS2SpriteLit.shader
- Assets/Borzblade/RetroRenderToolkit/Shaders/Sprites/PSXPS2SpriteUnlit.shader
- Assets/Borzblade/RetroRenderToolkit/Shaders/Outline/PSXPS2ScreenOutline.shader
- Assets/Borzblade/RetroRenderToolkit/Shaders/Outline/PSXPS2MaterialOutline.shader
- Assets/Borzblade/RetroRenderToolkit/Shaders/Core/PSXPS2RetroCommon.hlsl
- Assets/Borzblade/RetroRenderToolkit/Editor/Windows/RetroRenderToolkitWindow.cs
- Assets/Borzblade/RetroRenderToolkit/Editor/Tools/PSXPS2RetroInstaller.cs
- Assets/Borzblade/RetroRenderToolkit/Editor/Tools/RetroMaterialConverter.cs
- Assets/Borzblade/RetroRenderToolkit/Editor/Tools/RetroRenderToolkitMaterialFactory.cs
- Assets/Borzblade/RetroRenderToolkit/Editor/Tools/RetroMaterialPreset.cs
- Assets/Borzblade/RetroRenderToolkit/Editor/Tools/RetroSnapAnchorBaker.cs
- Assets/Borzblade/RetroRenderToolkit/Editor/ShaderGUI/PSXPS2SpriteShaderGUI.cs

Final Checklist

Before shipping or testing a scene:

1. URP is active.
2. The renderer feature is installed on the renderer used by the camera.
3. PSXPS2_GlobalPass.mat exists.
4. Default or custom toolkit materials exist.
5. Main opaque meshes use Hybrid Lit.
6. Terrain uses Terrain Lit as the material/template.
7. Leaves, grass, hair cards, fences, and grates use cutout or foliage shaders.
8. Alpha-card shadows have been checked.
9. Water planes use the water shader and have been checked for transparency sorting.
10. SpriteRenderers use the sprite shaders when sprite tint, vertex color, atlas behavior, or sorting layers matter.
11. Screen or material outlines are only enabled where they improve readability.
12. The renderer preset has been tuned for the scene, including fixed vertical resolution if used.
13. Depth fog is used for scene fog, and material fog is only enabled where a specific shader needs local stylized fog.

14. Strong vertex snap has been checked for edge gaps, with Snap Seam Reduction or baked snap anchors used only where needed.
15. Foliage Jiggle Amplitude is tuned near the camera and does not slide cards sideways.
16. Mobile builds use the mobile presets or reduced dither, scanline, bleed, outline, and depth-fade settings.